▶ The magnetic field induced by a current-carrying coil wound around a toroid is confined entirely to the toroid volume:

$$\mathbf{H} = \begin{cases} 0 & \text{for } r < a, \\ -\hat{\boldsymbol{\phi}}\,\frac{NI}{2\pi r} & \text{for } a < r < b, \\ 0 & \text{for } r > b. \end{cases}$$

Example 5-6: Magnetic Field of an Infinite Current Sheet

The x–y plane contains an infinite current sheet with surface current density $\mathbf{J}_s = \hat{\mathbf{x}}J_s$ (**Fig. 5-19**). Find the magnetic field $\mathbf{H}$ everywhere in space.

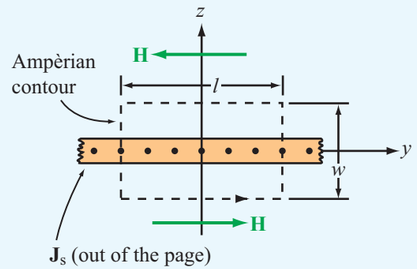


Figure 5-19 A thin current sheet in the x–y plane carrying a surface current density $\mathbf{J}_s = \hat{\mathbf{x}}J_s$ (Example 5-6).

Solution: From symmetry considerations and the right-hand rule, for $z > 0$ and $z < 0$ $\mathbf{H}$ must be in the directions shown in the figure. That is,

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}}H & \text{for } z > 0, \\ \hat{\mathbf{y}}H & \text{for } z < 0. \end{cases}$$

To evaluate the line integral in Ampère's law, we choose a rectangular Ampèrian path around the sheet with dimensions l and w (**Fig. 5-19**). Recalling that J_s represents current per unit length along the y direction, the total current crossing the surface of the rectangular loop is $I = J_s l$. Hence, applying Ampère's law over the loop in a counterclockwise direction, while noting that $\mathbf{H}$ is perpendicular to the paths of length w, we have

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = 2Hl = J_s l,$$

from which we obtain the result

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}}\,\frac{J_s}{2} & \text{for } z > 0, \\ \hat{\mathbf{y}}\,\frac{J_s}{2} & \text{for } z < 0. \end{cases} \tag{5.51}$$

Concept Question 5-8: What are the fundamental differences between electric and magnetic fields?

Concept Question 5-9: If the line integral of $\mathbf{H}$ over a closed contour is zero, does it follow that $\mathbf{H} = 0$ at every point on the contour? If not, what then does it imply?

Concept Question 5-10: Compare the utility of applying the Biot–Savart law versus applying Ampère's law for computing the magnetic field due to current-carrying conductors.

Concept Question 5-11: What is a toroid? What is the magnetic field outside the toroid?

5-4 Vector Magnetic Potential

In our treatment of electrostatic fields in Chapter 4, we defined the electrostatic potential V as the line integral of the electric field $\mathbf{E}$ and found that V and $\mathbf{E}$ are related by $\mathbf{E} = -\nabla V$. This relationship proved useful not only in relating electric field distributions in circuit elements (such as resistors and capacitors) to the voltages across them but also to determine $\mathbf{E}$ for a given charge distribution by first computing V using Eq. (4.48). We now explore a similar approach in connection with the magnetic flux density $\mathbf{B}$.

According to Eq. (5.44), $\nabla \cdot \mathbf{B} = 0$. We wish to define $\mathbf{B}$ in terms of a magnetic potential with the constraint that such a definition guarantees that the divergence of $\mathbf{B}$ is always zero. This can be realized by taking advantage of the vector identity given by Eq. (3.106b), which states that, for any vector $\mathbf{A}$,

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0. \tag{5.52}$$

Hence, by introducing the ***vector magnetic potential*** $\mathbf{A}$ such that

$$\mathbf{B} = \nabla \times \mathbf{A} \qquad (\text{Wb/m}^2), \tag{5.53}$$

we are guaranteed that $\nabla \cdot \mathbf{B} = 0$. The SI unit for $\mathbf{B}$ is the tesla (T). An equivalent unit is webers per square meter (Wb/m^2). Consequently, the SI unit for $\mathbf{A}$ is (Wb/m).

▶ 缠绕在环形线圈上的载流线圈感应的磁场完全限制在环形线圈体积内：

$$\mathbf{H} = \begin{cases} 0 & \text{for } r < a, \\ -\hat{\boldsymbol{\phi}}\,\frac{NI}{2\pi r} & \text{for } a < r < b, \\ 0 & \text{for } r > b. \end{cases}$$

例 5-6： 无限电流片的磁场

x-y 平面包含表面电流密度为 $J_s = \hat{x}J_s$ (的无限电流片。 5-19)。求空间中各处的磁场H。

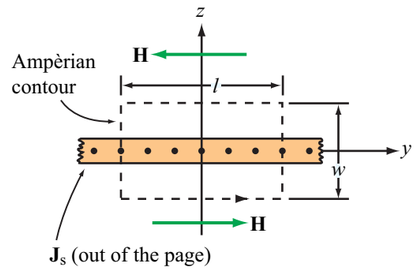


图 5-19 x-y 平面上的薄电流片，其表面电流密度为 J $_s = \hat{x}J_s$ (示例 5-6)。

解：从对称性考虑和右手定则，对于 z > 0 和 z < 0 H 必须处于图中所示的方向。那是，

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}}H & \text{for } z > 0, \\ \hat{\mathbf{y}}H & \text{for } z < 0. \end{cases}$$

评估 Amp 中的线路积分根据这条定律，我们选择一个矩形放大器尺寸为 l 和 w 的板材周围的埃里安路径（图 5-19）。回想起J_s表示沿 y 方向单位长度的电流，穿过矩形环路表面的总电流为 I=J_s湖因此，应用放大器逆时针方向的环路遵循埃雷定律，同时注意到 H 垂直于长度为 w 的路径，我们有

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = 2Hl = J_s l,$$

从中我们得到结果

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}}\,\frac{J_s}{2} & \text{for } z > 0, \\ \hat{\mathbf{y}}\,\frac{J_s}{2} & \text{for } z < 0. \end{cases} \tag{5.51}$$

概念问题 5-8：电场和磁场之间的根本区别是什么？

概念问题 5-9：如果 H 在闭合轮廓上的线积分为零，是否可以得出轮廓上每个点的 H = 0？如果不是，那么这意味着什么？

概念问题 5-10：比较应用毕奥-萨伐尔定律与应用安培环路定律计算载流导体产生的磁场的效用。

概念问题 5-11：什么是环形线圈？环形线圈外的磁场是多少？

5-4 矢量磁势

在第 4 章静电场的处理中，我们将静电势 V 定义为电场 E 的线积分，并发现 V 和 E 通过 E = −∇V 相关。事实证明，这种关系不仅有助于将电路元件（例如电阻器和电容器）中的电场分布与它们两端的电压联系起来，而且还可以通过首先使用方程式计算 V 来确定给定电荷分布的 E 。 （4.48）。我们现在探索与磁通密度 B 相关的类似方法。

根据方程。 (5.44), ∇ · B = 0。我们希望用磁势来定义 B ，并限制这样的定义保证 B 的散度始终为零。这可以通过利用等式给出的向量恒等式来实现。 (3.106b)，它指出，对于任何向量 A，

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0. \tag{5.52}$$

因此，通过引入矢量磁势 A 使得

$$\mathbf{B} = \nabla \times \mathbf{A} \qquad (\text{Wb/m}^2), \tag{5.53}$$

我们保证 ∇ · B = 0。B 的 SI 单位是特斯拉 (T)。等效单位是韦伯每平方米 (Wb/m^2)。因此，A 的 SI 单位为 (Wb/m) 。

Technology Brief 10: Electromagnets

William Sturgeon developed the first practical ***electromagnet*** in the 1820s. Today, the principle of the electromagnet is used in motors, relay switches in read/write heads for hard disks and tape drives, loud speakers, magnetic levitation, and many other applications.

Basic Principle

Electromagnets can be constructed in various shapes, including the linear ***solenoid*** and ***horseshoe*** geometries depicted in **Fig. TF10-1**. In both cases, when an electric current flows through the insulated wire coiled around the central core, it induces a magnetic field with lines resembling those generated by a bar magnet. The strength of the magnetic field is proportional to the current, the number of turns, and the magnetic permeability of the core material. By using a ***ferromagnetic core***, the field strength can be increased by several orders of magnitude, depending on the purity of the iron material. When subjected to a magnetic field, ferromagnetic materials, such as iron or nickel, get magnetized and act like magnets themselves.

Magnetic Relays

A magnetic relay is a ***switch*** or circuit breaker that can be activated into the "ON" and "OFF" positions magnetically. One example is the low-power ***reed relay*** used in telephone equipment, which consists of two flat nickel–iron blades separated by a small gap (**Fig. TF10-2**). The blades are shaped in such a way that, in the absence of an external force, they remain apart and unconnected (OFF position). Electrical contact between the blades

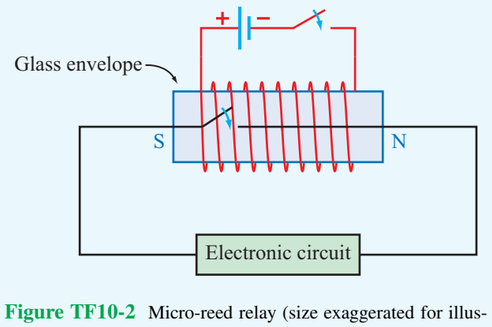
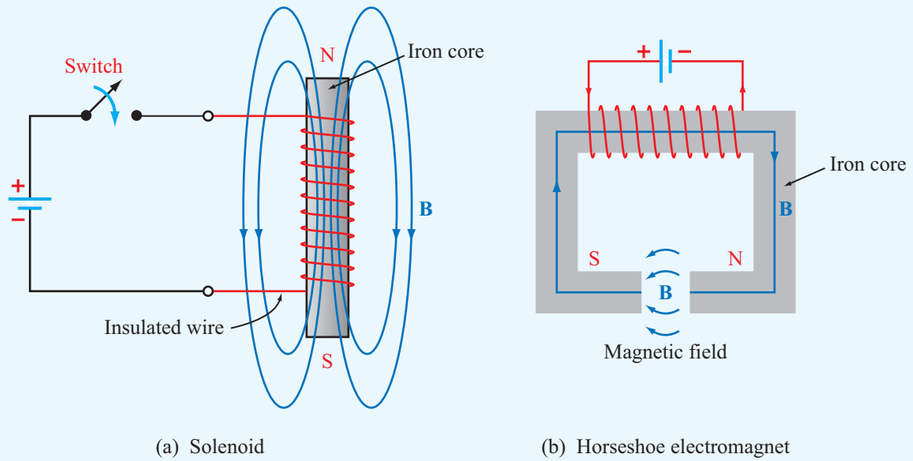


Figure TF10-2 Micro-reed relay (size exaggerated for illustration purposes).



(a) Solenoid (b) Horseshoe electromagnet

Figure TF10-1 Solenoid and horseshoe magnets.

技术简介 10：电磁铁

William Sturgeon 在 1820 年代开发出第一个实用的电磁体。如今，电磁体原理已用于电机、硬盘和磁带驱动器读/写头中的继电器开关、扬声器、磁悬浮以及许多其他应用。

基本原理

电磁体可以构造成各种形状，包括图 TF10-1 中所示的线性螺线管和马蹄形几何形状。在这两种情况下，当电流流过缠绕在中心磁芯周围的绝缘线时，它会感应出一个磁场，其磁场线类似于条形磁铁产生的磁场线。磁场强度与电流、匝数和磁芯材料的磁导率成正比。通过使用铁磁芯，磁场强度可以增加几个数量级，具体取决于铁材料的纯度。当受到磁场作用时，铁磁材料（例如铁或镍）会被磁化，并且本身就像磁铁一样。

磁性继电器

磁继电器是一种开关或断路器，可以通过磁力激活到“ON”和“OFF”位置。一个例子是电话设备中使用的低功率簧片继电器，它由两个由小间隙隔开的扁平镍铁刀片组成（图 TF10-2）。叶片的形状使得在没有外力的情况下，它们保持分开且不连接（关闭位置）。刀片之间的电接触

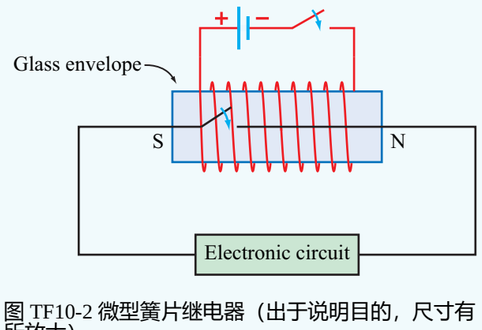
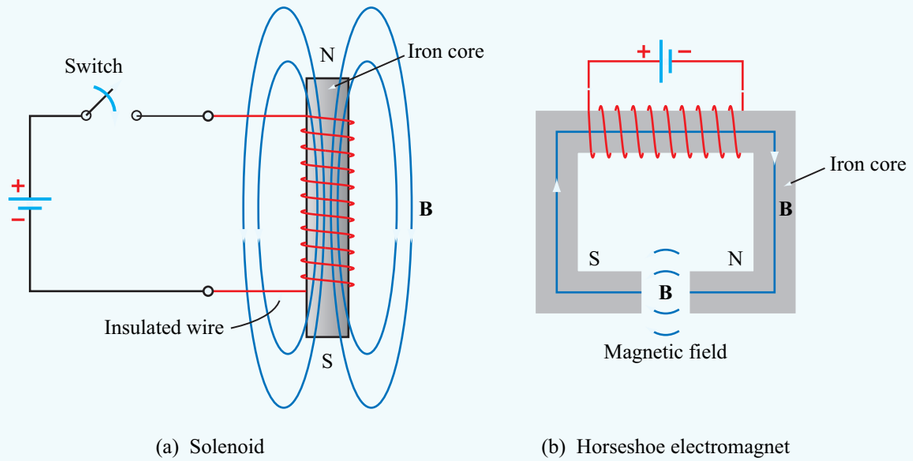


图 TF10-2 微型簧片继电器（出于说明目的，尺寸有所放大）。



(a) Solenoid (b) Horseshoe electromagnet

图 TF10-1 螺线管和马蹄形磁铁。

(ON position) is realized by applying a magnetic field along their length. The field, induced by a current flowing in the wire coiled around the glass envelope, causes the two blades to assume opposite magnetic polarities, thereby forcing them to attract together and close out the gap.

The Doorbell

In a doorbell circuit (**Fig. TF10-3**), the doorbell button is a switch; pushing and holding it down serves to connect the circuit to the household ac source through an appropriate ***step-down transformer***. The current from the source flows through the electromagnet, via a contact arm with only one end anchored in place (and the other moveable), and onward to the switch. The magnetic field generated by the current flowing in the windings of the electromagnet pulls the unanchored end of the contact arm (which has an iron bar on it) closer in—in the direction of the electromagnet—thereby losing connection with the metal contact and severing current flow in the circuit. With no magnetic field to pull on the contact arm, it snaps back into its earlier position, re-establishing the current in the circuit. This back-and-forth cycle is repeated many times per second, so long as the doorbell button continues to be pushed down, and with every cycle, the clapper arm attached to the contact arm hits the metal bell and generates a ringing sound.

The Loudspeaker

By using a combination of a stationary, permanent magnet and a moveable electromagnet, the electromagnet/speaker-cone of the loudspeaker (**Fig. TF10-4**) can be made to move back and forth in response to the electrical

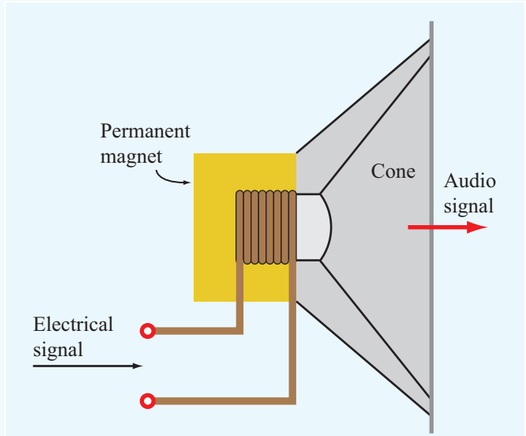


Figure TF10-4 The basic structure of a speaker.

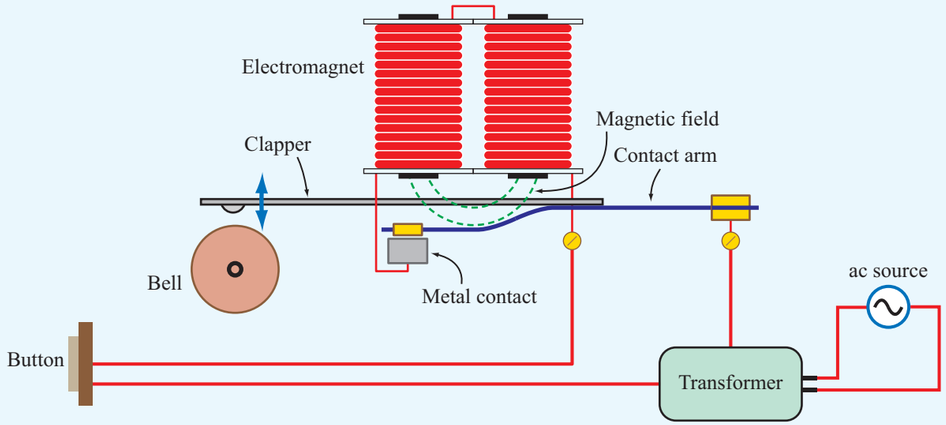


Figure TF10-3 Basic elements of a doorbell.

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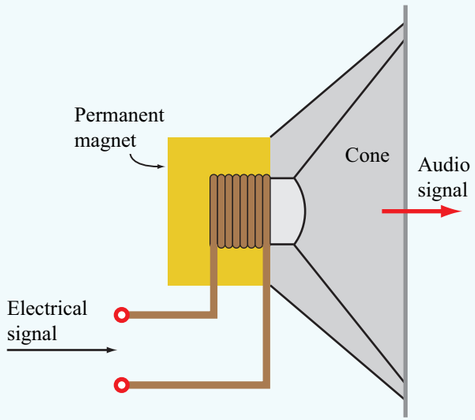
(ON 位置)是通过沿其长度施加磁场来实现的。由缠绕在玻璃外壳上的导线中流动的电流所产生的磁场，使两个叶片呈现相反的磁极性，从而迫使它们吸引在一起并关闭间隙。

门铃

在门铃电路中（图TF10-3），门铃按钮是一个开关；按下并按住它可以通过适当的降压变压器将电路连接到家用交流电源。来自源头的电流流过电磁体，通过只有一端固定到位（另一端可移动）的接触臂，然后流向开关。电磁体绕组中流动的电流产生的磁场将接触臂的非锚定端（上面有一个铁棒）拉近（沿电磁体的方向），从而失去与金属触点的连接并切断电路中的电流。由于没有磁场拉动接触臂，它会弹回原来的位置，从而在电路中重新建立电流。这种来回循环每秒重复多次，只要继续按下门铃按钮，每循环一次，连接到接触臂上的拍板臂就会撞击金属铃并产生铃声。

扩音器

通过使用固定的永磁体和可移动电磁体的组合，扬声器的电磁体/扬声器锥体（图 TF10-4）可以根据电力来前后移动。



图TF10-4 扬声器的基本结构。

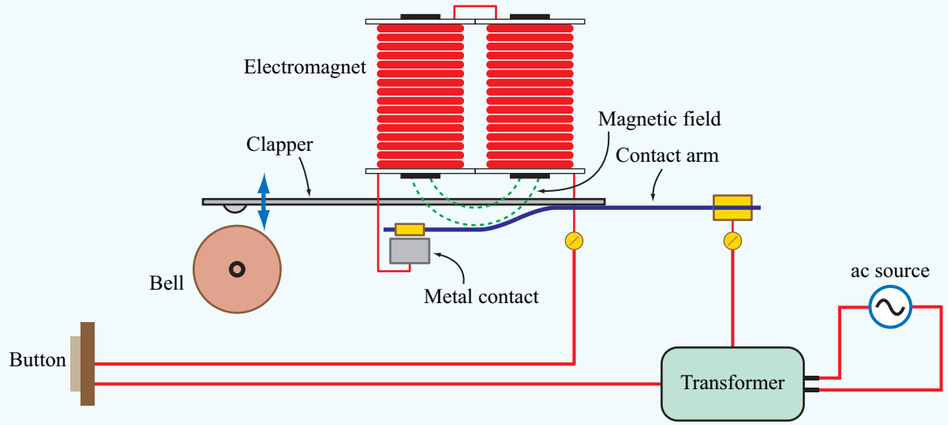
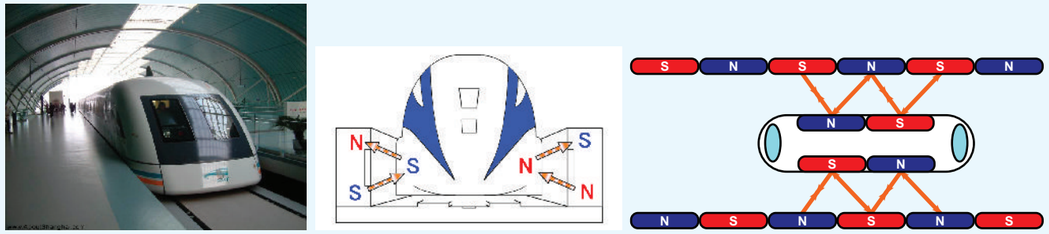


图 TF10-3 门铃的基本元件。



(a) (b) (c)

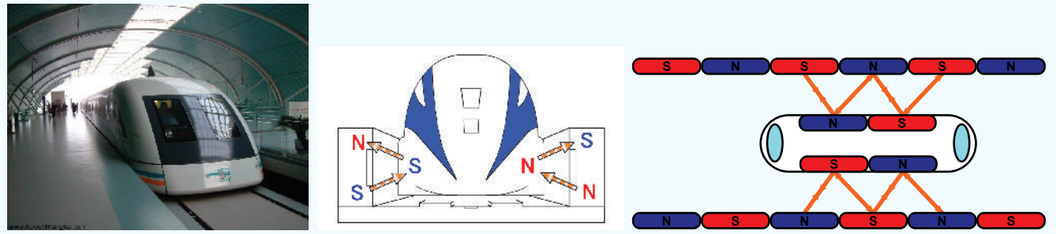
Figure TF10-5 (a) A maglev train, (b) electrodynamic suspension of an SCMaglev train, and (c) electrodynamic maglev propulsion via propulsion coils.

signal exciting the electromagnet. The vibrating movement of the cone generates sound waves with the same distribution of frequencies as contained in the spectrum of the electrical signal.

Magnetic Levitation

► Magnetically levitated trains (**Fig. TF10-5**(a)), called ***maglevs*** for short, can achieve speeds as high as 500 km/hr, primarily because there is no friction between the train and the track. ◄

The train basically floats at a height of 1 or more centimeters above the track, which is made possible by magnetic levitation (**Fig. TF10-5**(b)). The train carries superconducting electromagnets that induce currents in coils built into the guide rails alongside the train. The magnetic interaction between the train's superconducting electromagnets and the guide-rail coils serves not only to levitate the train but also to propel it along the track.



(a) (b) (c)

图 TF10-5 (a) 磁悬浮列车；(b) 超导磁悬浮列车的电动悬浮；(c) 利用推进线圈实现电动磁悬浮推进。

信号激励电磁铁。锥体的振动运动产生与电信号频谱中包含的频率分布相同的声波。

磁悬浮

► 磁悬浮列车（图TF10-5（a））， 简称磁悬浮，可以达到高达500公里/小时的速度，主要是因为列车与轨道之间没有摩擦。 ◄

火车基本上漂浮在轨道上方 1 厘米或以上的高度，这是通过磁悬浮实现的（图 TF10-5（b））。火车装有超导电磁体，可在火车旁边内置于导轨中的线圈中感应电流。火车超导电磁体和导轨线圈之间的磁相互作用不仅可以使火车悬浮，还可以沿着轨道推动火车。

5-4.1 Vector Poisson's Equation

With $\mathbf{B} = \mu\mathbf{H}$, the differential form of Ampère's law given by Eq. (5.46) can be written as

$$\nabla \times \mathbf{B} = \mu \mathbf{J}. \tag{5.54}$$

If we substitute Eq. (5.53) into Eq. (5.54), we obtain

$$\nabla \times (\nabla \times \mathbf{A}) = \mu \mathbf{J}. \tag{5.55}$$

For any vector $\mathbf{A}$, the Laplacian of $\mathbf{A}$ obeys the vector identity given by Eq. (3.113), that is,

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A}), \tag{5.56}$$

where, by definition, $\nabla^2\mathbf{A}$ in Cartesian coordinates is

$$\begin{aligned} \nabla^2 \mathbf{A} &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\mathbf{A} \\ &= \hat{\mathbf{x}}\nabla^2 A_x + \hat{\mathbf{y}}\nabla^2 A_y + \hat{\mathbf{z}}\nabla^2 A_z. \end{aligned} \tag{5.57}$$

Combining Eq. (5.55) with Eq. (5.56) gives

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu \mathbf{J}. \tag{5.58}$$

This equation contains a term involving $\nabla \cdot \mathbf{A}$. It turns out that we have a fair amount of latitude in specifying a value or a mathematical form for $\nabla \cdot \mathbf{A}$ without conflicting with the requirement represented by Eq. (5.53). The simplest among these allowed restrictions on $\mathbf{A}$ is

$$\nabla \cdot \mathbf{A} = 0. \tag{5.59}$$

Using this choice in Eq. (5.58) leads to the ***vector Poisson's equation***

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}. \tag{5.60}$$

Using the definition for $\nabla^2\mathbf{A}$ given by Eq. (5.57), the vector Poisson's equation can be decomposed into three scalar Poisson's equations:

$$\nabla^2 A_x = -\mu J_x, \tag{5.61a}$$
$$\nabla^2 A_y = -\mu J_y, \tag{5.61b}$$
$$\nabla^2 A_z = -\mu J_z. \tag{5.61c}$$

In electrostatics, Poisson's equation for the scalar potential V is given by Eq. (4.70) as

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon}, \tag{5.62}$$

and its solution for a volume charge distribution ρ_v occupying a volume $\mathcal{V}'$ was given by Eq. (4.71) as

$$V = \frac{1}{4\pi\varepsilon}\int_{\mathcal{V}'} \frac{\rho_v}{R'}\, d\mathcal{V}'. \tag{5.63}$$

Poisson's equations for A_x, A_y, and A_z are mathematically identical in form to Eq. (5.62). Hence, for a current density $\mathbf{J}$ with x component J_x distributed over a volume $\mathcal{V}'$, the solution for Eq. (5.61a) is

$$A_x = \frac{\mu}{4\pi}\int_{\mathcal{V}'} \frac{J_x}{R'}\, d\mathcal{V}' \qquad \text{(Wb/m)}. \tag{5.64}$$

Similar solutions can be written for A_y in terms of J_y and for A_z in terms of J_z. The three solutions can be combined into a vector equation:

$$\mathbf{A} = \frac{\mu}{4\pi}\int_{\mathcal{V}'} \frac{\mathbf{J}}{R'}\, d\mathcal{V}' \qquad \text{(Wb/m)}. \tag{5.65}$$

In view of Eq. (5.23), if the current distribution is specified over a surface S', then $\mathbf{J}\, d\mathcal{V}'$ should be replaced with $\mathbf{J}_s\, ds'$ and $\mathcal{V}'$ should be replaced with S'; similarly, for a line distribution, $\mathbf{J}\, d\mathcal{V}'$ should be replaced with $I\, d\mathbf{l}'$, and the integration should be performed over the associated path l'.

▶ The vector magnetic potential provides a third approach for computing the magnetic field due to current-carrying conductors in addition to the methods suggested by the Biot–Savart and Ampère laws. ◀

For a specified current distribution, Eq. (5.65) can be used to find $\mathbf{A}$, and then Eq. (5.53) can be used to find $\mathbf{B}$. Except for simple current distributions with symmetrical geometries that lend themselves to the application of Ampère's law, in practice we often use the approaches provided by the Biot–Savart law and the vector magnetic potential, and among these two, the latter often is more convenient to apply because it is easier to perform the integration in Eq. (5.65) than that in Eq. (5.22).

5-4.2 Magnetic Flux

The ***magnetic flux*** Φ linking a surface S is defined as the total magnetic flux density passing through it, or

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \qquad \text{(Wb)}. \tag{5.66}$$

If we insert Eq. (5.53) into Eq. (5.66) and then invoke Stokes's theorem, we obtain

$$\Phi = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l} \qquad \text{(Wb)}, \tag{5.67}$$

5-4.1矢量泊松方程

对于 B = μ H，Ampère 定律的微分形式由方程式给出。 (5.46) 可以写成

$$\nabla \times \mathbf{B} = \mu \mathbf{J}. \tag{5.54}$$

如果我们替换方程。 （5.53）代入等式。(5.54)，我们得到

$$\nabla \times (\nabla \times \mathbf{A}) = \mu \mathbf{J}. \tag{5.55}$$

对于任何向量 A，A 的拉普拉斯算子服从式（1）给出的向量恒等式： （3.113），即

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A}), \tag{5.56}$$

其中，根据定义，笛卡尔坐标中的 ∇^2A 为

$$\begin{aligned} \nabla^2 \mathbf{A} &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\mathbf{A} \\ &= \hat{\mathbf{x}}\nabla^2 A_x + \hat{\mathbf{y}}\nabla^2 A_y + \hat{\mathbf{z}}\nabla^2 A_z. \end{aligned} \tag{5.57}$$

结合方程。 (5.55) 与等式。 (5.56) 给出

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu \mathbf{J}. \tag{5.58}$$

这个方程包含一个涉及 ∇· A 的项。事实证明，我们在为 ∇· A 指定一个值或数学形式方面有相当大的自由度，而不与方程 1 所表示的要求相冲突。 （5.53）。对 A 的这些允许的限制中最简单的是

$$\nabla \cdot \mathbf{A} = 0. \tag{5.59}$$

在方程式中使用这个选择。 (5.58) 导出矢量泊松方程

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}. \tag{5.60}$$

使用方程式给出的 ∇^2A 的定义。 （5.57）式中，矢量泊松方程可以分解为三个标量泊松方程：

$$\nabla^2 A_x = -\mu J_x, \tag{5.61a}$$
$$\nabla^2 A_y = -\mu J_y, \tag{5.61b}$$
$$\nabla^2 A_z = -\mu J_z. \tag{5.61c}$$

在静电学中，标量势 V 的泊松方程由方程 1 给出。 (4.70) 作为

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon}, \tag{5.62}$$

其占据体积 υ′ 的体积电荷分布 ρ_v 的解由式 (1) 给出。 (4.71) 作为

$$V = \frac{1}{4\pi\varepsilon}\int_{\upsilon'} \frac{\rho_v}{R'}\, d\upsilon'. \tag{5.63}$$

A_x、A_y 和 A_z 的泊松方程在数学上与方程 1 相同。 （5.62）。因此，对于 x 分量 J_x 分布在体积 υ′ 上的电流密度 J，方程 1 的解为： (5.61a) 是

$$A_x = \frac{\mu}{4\pi}\int_{\upsilon'} \frac{J_x}{R'}\, d\upsilon' \qquad \text{(Wb/m)}. \tag{5.64}$$

对于 A_y 可以根据 J_y 编写类似的解决方案，对于 A_z 可以根据 J_z 编写类似的解决方案。这三个解可以组合成一个向量方程：

$$\mathbf{A} = \frac{\mu}{4\pi}\int_{\upsilon'} \frac{\mathbf{J}}{R'}\, d\upsilon' \qquad \text{(Wb/m)}. \tag{5.65}$$

鉴于方程。 (5.23)，如果在表面 S′ 上指定电流分布，则 J dυ′ 应替换为 J_s ds′，υ′ 应替换为 S′；类似地，对于线分布，J dυ′ 应替换为 I dl′，并且积分应在相关路径 l′ 上执行。

▶除了 Biot-Savart 和 Amp 建议的方法之外，矢量磁势提供了第三种计算载流导体产生的磁场的方法这是法律。◀

对于特定的电流分布，方程。 (5.65) 可用于求 A，然后可得方程 (5.65)。式（5.53）可以用来求B。除了具有对称几何形状的简单电流分布适合应用安培环路定律之外，在实践中我们经常使用毕奥-萨伐尔定律和矢量磁势提供的方法，而在这两者中，后者通常更方便应用，因为更容易执行式（5.53）中的积分。 (5.65) 比等式中的值（5.22）。

5-4.2 磁通量

连接表面 S 的磁通量 Φ 定义为穿过该表面的总磁通量密度，或者

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \qquad \text{(Wb)}. \tag{5.66}$$

如果我们插入方程。 （5.53）代入等式。 (5.66) 然后调用斯托克斯定理，我们得到

$$\Phi = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l} \qquad \text{(WB)}, \tag{5.67}$$